

Philadelphia Restaurant Yelp Optimization Challenge

1 Business Context

You are working with a Philadelphia-based restaurant that wants to improve its Yelp rating and overall customer perception in an increasingly competitive local dining market. Yelp ratings influence customer foot traffic, online visibility and search ranking, and long-term revenue and business survival. Therefore, maximizing Yelp ratings is important for any restaurant, and you have been hired to provide this restaurant with insights that they can use to help improve their Yelp rating.

2 Your Task

Using Yelp data for **Philadelphia businesses**, analyze which factors are most strongly associated with higher Yelp review ratings and provide **actionable recommendations** that a restaurant owner could realistically implement. You may explore questions such as:

- Which business attributes are most predictive of higher review stars?
- Does location play a role in ratings?
- How do reviewer characteristics influence ratings?
- Are certain amenities more valuable for specific types of restaurants?
- How do recent reviews differ from older ones?

Your final deliverable should translate data insights into **clear business recommendations** in a 5-10 minute presentation.

3 Data Overview

3.1 Data Source and Structure

All data for this challenge comes from the **Yelp Open Dataset**, a publicly available dataset provided by Yelp for academic and educational use. For the purposes of this Datathon, the data has been filtered to include **only businesses, reviews, and users associated with the city of Philadelphia, PA**.

Rather than a single file, the dataset is provided as **five separate CSV files**. Participants are expected to join these datasets as needed using unique identifiers such as **business_id** and

`user_id`. This structure mirrors real-world data environments and allows teams flexibility in how they construct their analytical dataset.

Note: Users are only included if they have written at least one review or tip for a Philadelphia business.

4 Files Included in the Dataset

4.1 Business Data (`business.csv`)

This file contains core business-level information including location, categories, and attributes. Each row represents a unique business.

Column	Description
<code>business_id</code>	Unique Yelp business identifier
<code>name</code>	Business name
<code>address, city, state, postal_code</code>	Business location fields
<code>latitude, longitude</code>	Geographic coordinates
<code>stars</code>	Overall Yelp star rating (rounded to half-stars)
<code>review_count</code>	Total number of Yelp reviews for the business
<code>is_open</code>	Indicator for whether the business is currently open
<code>attributes</code>	Nested JSON object of business attributes (e.g., parking, takeout, reservations)
<code>categories</code>	Comma-separated list of Yelp categories
<code>hours</code>	Business hours by day (24-hour format)

Participants should use the `categories` field to filter the dataset to **restaurants only**. This decision is intentionally left open-ended. Participants, for example, may decide to specifically focus on cafes, or certain kinds of cuisines.

4.2 Review Data (`review.csv`)

This file contains full Yelp reviews. Each row represents a single review written by a user for a business.

Column	Description
<code>review_id</code>	Unique review identifier
<code>user_id</code>	Identifier for the user who wrote the review
<code>business_id</code>	Identifier for the business being reviewed
<code>stars</code>	Star rating (1–5)
<code>date</code>	Review date (YYYY-MM-DD)
<code>text</code>	Full review text

useful, funny, cool	Engagement metrics for the review
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4.3 User Data (user.csv)

This file contains metadata about Yelp users who have written at least one review or tip for a Philadelphia business.

Column	Description
user_id	Unique user identifier
name	User's first name
review_count	Total number of reviews written by the user
yelping_since	Date the user joined Yelp
fans	Number of fans the user has
elite	Years the user was an elite Yelp reviewer
average_stars	User's average star rating across all reviews
useful, funny, cool	Total engagement metrics received by the user

The `friends` column is included for completeness but is not recommended for use, as many referenced users are not present in the dataset.

4.4 Tips Data (tip.csv)

Tips are short, quick comments written by users that provide concise insights or recommendations.

Column	Description
text	Tip text
date	Date the tip was written
compliment_count	Number of compliments received
business_id	Business associated with the tip
user_id	User who wrote the tip

4.5 Check-In Data (checkin.csv)

Check-ins represent instances where users publicly announce a visit to a business.

Column	Description
business_id	Business associated with the check-ins
date	Comma-separated list of timestamps for each check-in

5 Suggested Approach

There is no single correct way to approach this challenge. However, successful analyses often begin with a clear structure and hypothesis. Participants are encouraged to:

- Join multiple datasets using `business_id` and `user_id` to construct a review-level or business-level analytical table.
- Filter the data to include only restaurants using the `categories` field. Teams are welcome to focus on specific cuisines or restaurant types if doing so simplifies analysis.
- Begin with one or more hypotheses (e.g., “Restaurants offering takeout receive higher ratings” or “More experienced reviewers give harsher ratings” or “Cafes in general have higher ratings than restaurants”) and test whether the data supports them.
- Use exploratory analysis to refine questions before building models or visualizations.
- More advanced teams may choose to build linear or logistic regression models to quantify how specific features (e.g., amenities, location, review characteristics, or user experience) are associated with higher Yelp ratings.
- Models can also be used to compare restaurant types, assess the relative importance of different attributes, or explore how reviewer behavior influences observed ratings.

Clear reasoning, thoughtful assumptions, and interpretable insights are valued more than model complexity.

6 Use of AI and External Resources

Participants are explicitly allowed and encouraged to use:

- AI tools (including large language models)
- Online documentation and tutorials
- Publicly available articles or reports on Philadelphia’s restaurant scene

External context may help inform hypotheses, feature engineering, or interpretation of results. However, teams should ensure that any assumptions or external insights are clearly explained.

7 Expected Output

Teams should deliver a **5–10 minute presentation** that clearly communicates their analytical process and conclusions. Presentations should introduce the primary question or hypothesis explored, explain how the data was used to investigate that question, and summarize the key findings. Analyses should translate into **clear, actionable, data-backed recommendations** that a restaurant owner could realistically implement. Supporting tables, charts, or visualizations should be used where appropriate to reinforce insights and improve clarity.